The Foreign Exchange Market

Reference: The Economics of Money, Banking, and Financial Markets by Mishkin

Indraprastha Institute of Information Technology, Delhi

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Foreign Exchange Market (1 of 2)

- **Exchange rate**: price of one currency in terms of another
- **Foreign exchange market**: the financial market where exchange rates are determined
- **Spot transaction**: immediate (two-day) exchange of bank deposits
  - Spot exchange rate
- **Forward transaction**: the exchange of bank deposits at some specified future date
  - Forward exchange rate

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Foreign Exchange Market (2 of 2)

- **Appreciation**: a currency rises in value relative to another currency
- **Depreciation**: a currency falls in value relative to another currency
- When a country's currency appreciates, the country's goods become more expensive to foreigners and foreign goods in that country become less expensive to domestic economic agents.

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Why are exchange rates so important?

- Exchange rates are important because they affect the relative price of domestic and foreign goods.
- The dollar price of French goods to an American is determined by the interaction of two factors:
  - the price of French goods in euros
  - and the euro/dollar exchange rate

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Why are exchange rates so important?

- If the price of the wine in France is 1,000 euros and the exchange rate is $1.08 to the euro, the wine will cost Wanda (an American) $1,080 (=1,000 euros X $1.08/euro). Now suppose that
- Wanda delays her purchase by two months, at which time the euro has appreciated to $1.20 per euro.
- If the domestic price of the wine remains 1,000 euros, its dollar cost will have risen from $1,080 to $1,200.

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Why are exchange rates so important?

- The same currency appreciation, however, makes the price of foreign goods in that country less expensive.
- At an exchange rate of $1.08 per euro, a HP computer priced at $2,000 costs Pierre the Programmer in France 1,852 euros.
- If the exchange rate increases to $1.20 per euro, the computer will cost only 1,667 euros.

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Why are exchange rates so important?

- A depreciation of the euro lowers the cost of French goods in America but raises the cost of American goods in France.
- If the euro drops in value to $0.90, Wanda's wine bottle will cost her only $900 instead of $1,080,
- and the HP computer will cost Pierre 2,222 euros rather than 1,852.

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Why are exchange rates so important?

- When a country's currency appreciates (rises in value relative to other currencies), the country's goods abroad become more expensive and foreign goods in that country become cheaper (holding domestic prices constant in the two countries).
- Conversely, when a country's currency depreciates, its goods abroad become cheaper and foreign goods in that country become more expensive.
- Appreciation of a currency can make it harder for domestic manufacturers to sell their goods abroad and can increase competition at home from foreign goods, because they cost less.

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Exchange Rates in the Long Run

- Like the price of any good or asset in a free market, exchange rates are determined by the interaction of supply and demand.
- To simplify our analysis of exchange rates in a free market, we divide it into two parts.
- First, we examine how exchange rates are determined in the long run;
- then we use our knowledge of the long-run determinants of the exchange rate to help us understand how they are determined in the short run.

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Exchange Rates in the Long Run

- Law of one price: the prices of an identical good should be the same throughout the world if trade barriers are low
- Suppose that American steel costs $100 per ton and identical Japanese steel costs 10,000 yen per ton.
- For the law of one price to hold, the exchange rate between the yen and the dollar must be 100 yen per dollar ($0.01 per yen)
- so that one ton of American steel sells for 10,000 yen in Japan (the price of Japanese steel) and one ton of Japanese steel sells for $100 in the United States (the price of U.S. steel).

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Exchange Rates in the Long Run

- One of the most prominent theories of how exchange rates are determined is the theory of purchasing power parity (PPP).
- It states that exchange rates between any two currencies will adjust to reflect changes in the price levels of the two countries.
- The theory of PPP is simply an application of the law of one price to national price levels rather than to individual prices.

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Exchange Rates in the Long Run

- Why the Theory of Purchasing Power Parity Cannot Fully Explain Exchange Rates?
- The PPP conclusion that exchange rates are determined solely by changes in relative price levels rests on the assumption
- All goods are identical in both countries
- Transportation costs and trade barriers are very low.
- The assumption that goods are identical may not be too reasonable for American and Japanese products.

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Exchange Rates in the Long Run

- PPP theory furthermore does not take into account that many goods and services (whose prices are included in a measure of a country's price level) are not traded across borders.
- Housing, land, and services such as restaurant meals, haircuts, and golf lessons are not traded goods.
- So even though the prices of these items might rise and lead to a higher price level relative to another country's, there would be little direct effect on the exchange rate.

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Factors That Affect Exchange Rates in the Long Run

- The basic rule: Anything that increases the demand for domestically produced goods relative to foreign traded goods tends to appreciates domestic currency
  - Relative price levels
  - Trade barriers
  - Preferences for domestic versus foreign goods
  - Productivity

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Factors That Affect Exchange Rates in the Long Run

- Similarly, anything that increases the demand for foreign goods relative to domestic goods tends to depreciate the domestic currency
- Domestic goods will continue to sell well only if the value of the domestic currency is lower.

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Relative Price Levels

- When prices of American goods rise (holding prices of foreign goods constant), the demand for American goods falls and the dollar tends to depreciate so that American goods can still sell well.
- By contrast, if prices of Japanese goods rise so that the relative prices of American goods fall, the demand for American goods increases, and the dollar tends to appreciate, because American goods will continue to sell well even with a higher value of the domestic currency.
- In the long run, a rise in a country's price level (relative to the foreign price level) causes its currency to depreciate, and a fall in the country's relative price level causes its currency to appreciate.

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Trade Barriers

- Barriers to free trade such as tariffs (taxes on imported goods) and quotas (restrictions on the quantity of foreign goods that can be imported) can affect the exchange rate.
- Suppose that the United States increases its tariff or puts a lower quota on Japanese steel.
- These increases in trade barriers increase the demand for American steel, and the dollar tends to appreciate because American steel will still sell well even with a higher value of the dollar.
- Increasing trade barriers cause a country's currency to appreciate in the long run.

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Preferences for Domestic Versus Foreign Goods

- If the Japanese develop an appetite for American goods, the increased demand for American goods (exports) tends to appreciate the dollar.
- American goods will continue to sell well even at a higher value for the dollar.
- Likewise, if Americans decide that they prefer Japanese cars to American cars, the increased demand for Japanese goods (imports) tends to depreciate the dollar.
- Increased demand for a country's exports causes its currency to appreciate in the long run; conversely, increased demand for imports causes the domestic currency to depreciate.

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Productivity

- If one country becomes more productive than other countries, businesses in that country can lower the prices of domestic goods relative to foreign goods and still earn a profit.
- As a result, the demand for domestic goods rises, and the domestic currency tends to appreciate.
- If, however, its productivity lags behind that of other countries, its goods become relatively more expensive, and the currency tends to depreciate.
- In the long run, as a country becomes more productive relative to other countries, its currency appreciates

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Tip

- The trick to figuring out what long-run effect a factor has on the exchange rate is to remember the following:
- If a factor increases the demand for domestic goods relative to foreign goods, the domestic currency will appreciate
- and if a factor decreases the relative demand for domestic goods, the domestic currency will depreciate.

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Summary Table 1 Factors That Affect Exchange Rates in the Long Run

Factor	Change in Factor	Response of the Exchange Rate, *E**
Domestic price level†	↑	↓
Trade barriers†	↑	↑
Import demand	↑	↓
Export demand	↑	↑
Productivity†	↑	↑

*Units of foreign currency per dollar: ↑ indicates domestic currency appreciation; ↓, depreciation.

†Relative to other countries.

Note: Only increases (↑) in the factors are shown; the effects of decreases in the variables on the exchange rate are the opposite of those indicated in the "Response" column.

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Nominal vs. Real Exchange Rate

- Real exchange rate: the rate at which domestic goods can be exchanged for foreign goods
- For example, let us consider a basket of goods in New York worth 50 USD; meanwhile, the same basket of goods is worth 7500 JPY in Tokyo. The nominal exchange rate is 100 JPN/USD.
- The real exchange rate is "how many foreign goods you can buy by paying one unit of domestic foods"
- In the above question, the real rate is

$$E_R = 50*100/7500 = 0.66$$

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Exchange Rates in the Short Run: A Supply and Demand Analysis

- We have developed a theory of the long-run behavior of exchange rates.
- To understand why exchange rates exhibit such large changes from day to day, we must develop a theory of how current exchange rates (spot exchange rates) are determined in the short run.
- The key to understanding the short-run behavior of exchange rates is to recognize that an exchange rate is the price of domestic bank deposits (those denominated in the domestic currency) in terms of foreign bank deposits (those denominated in the foreign currency).

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Exchange Rates in the Short Run: A Supply and Demand Analysis

- Earlier approaches to exchange rate determination emphasized the role of import and export demand.
- The more modern asset market approach used here does not emphasize the flows of purchases of exports and imports over short periods, because these transactions are quite small relative to the amount of domestic and foreign bank deposits at any given time.
- Thus over short periods, decisions to hold domestic or foreign assets play a much greater role in exchange rate determination than the demand for exports and imports does.

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Equilibrium in the Foreign Exchange Market

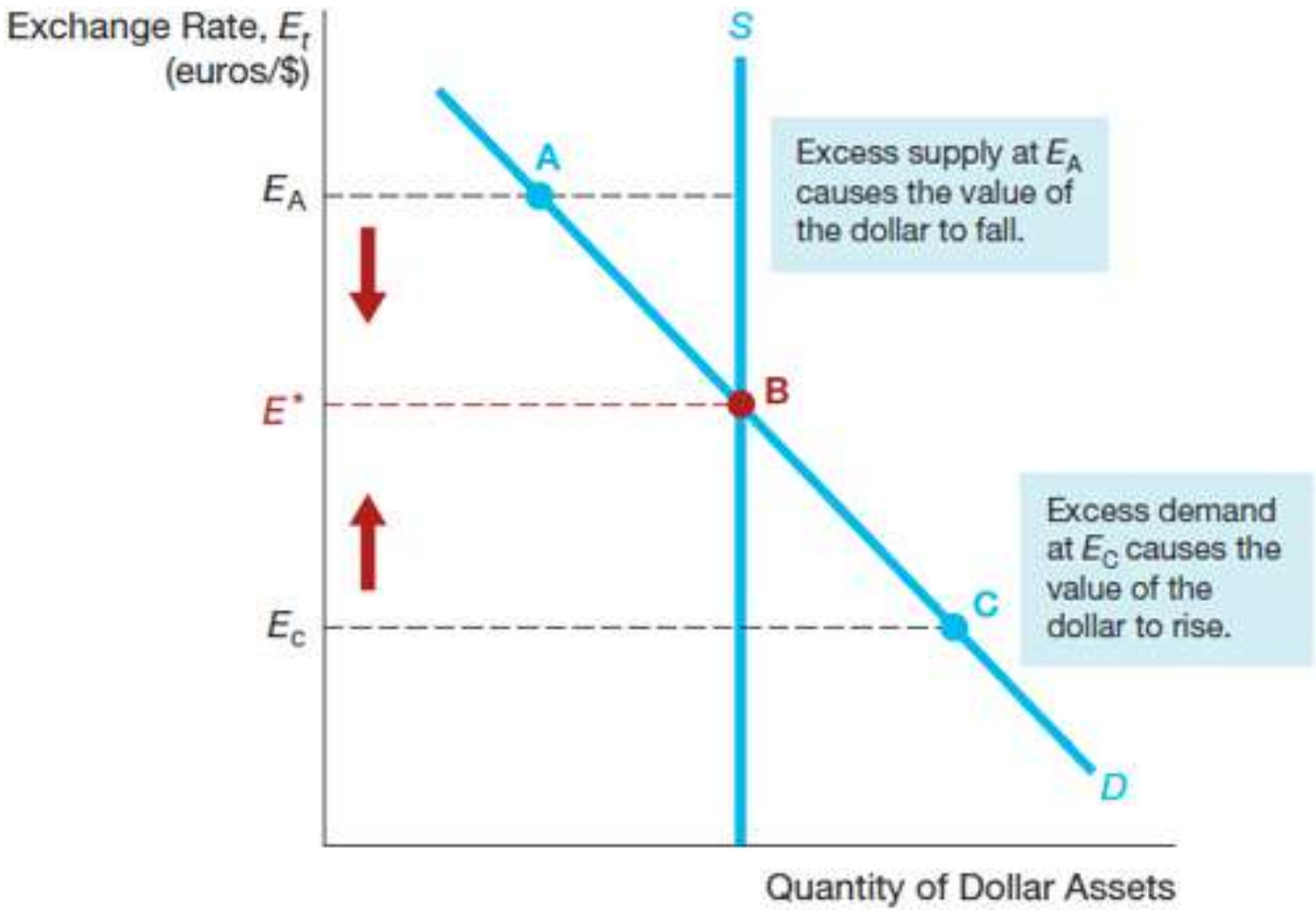

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Figure 3 Response to an Increase in the Domestic Interest Rate, i^D

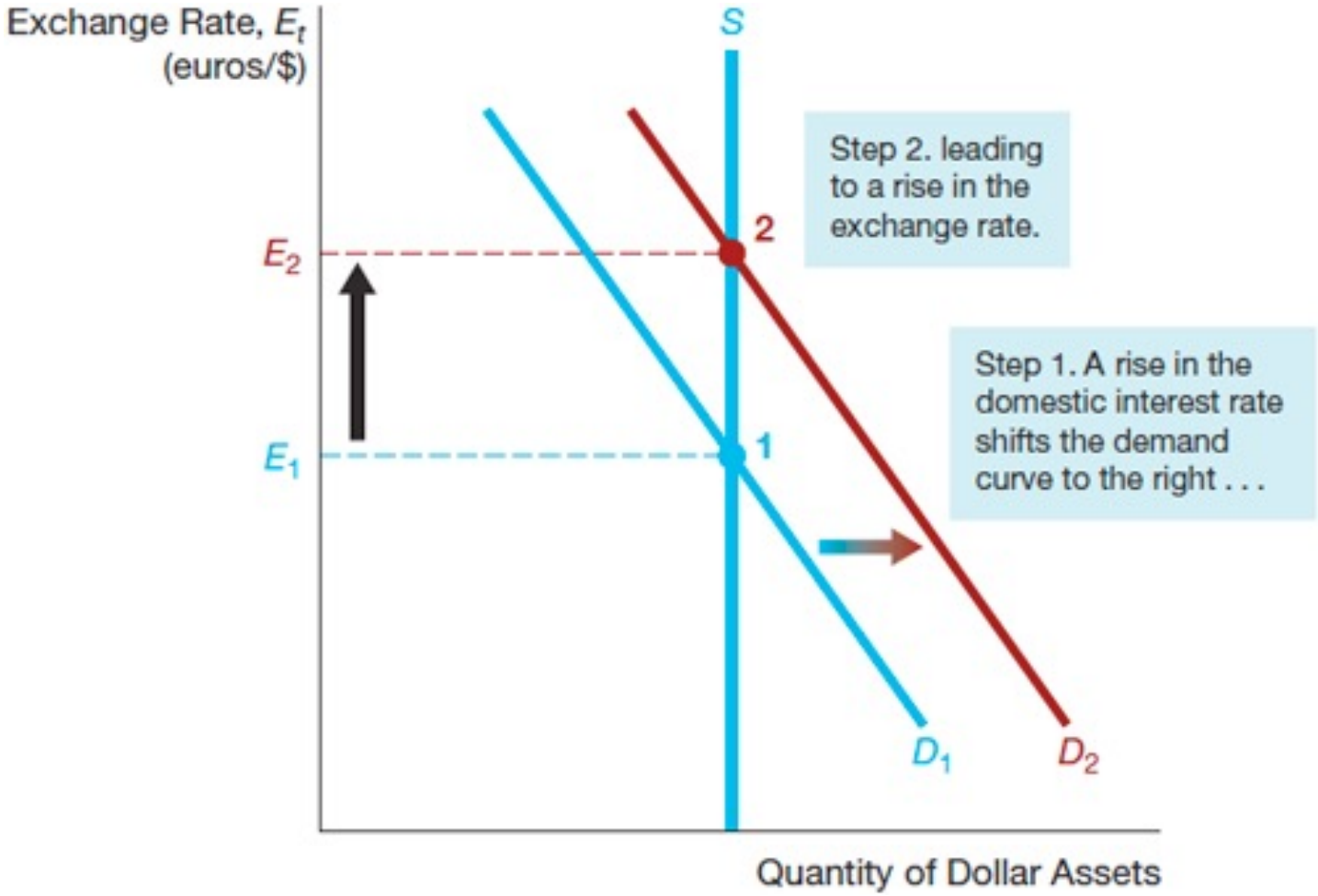

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Figure 4 Response to an Increase in the Foreign Interest Rate, i^F

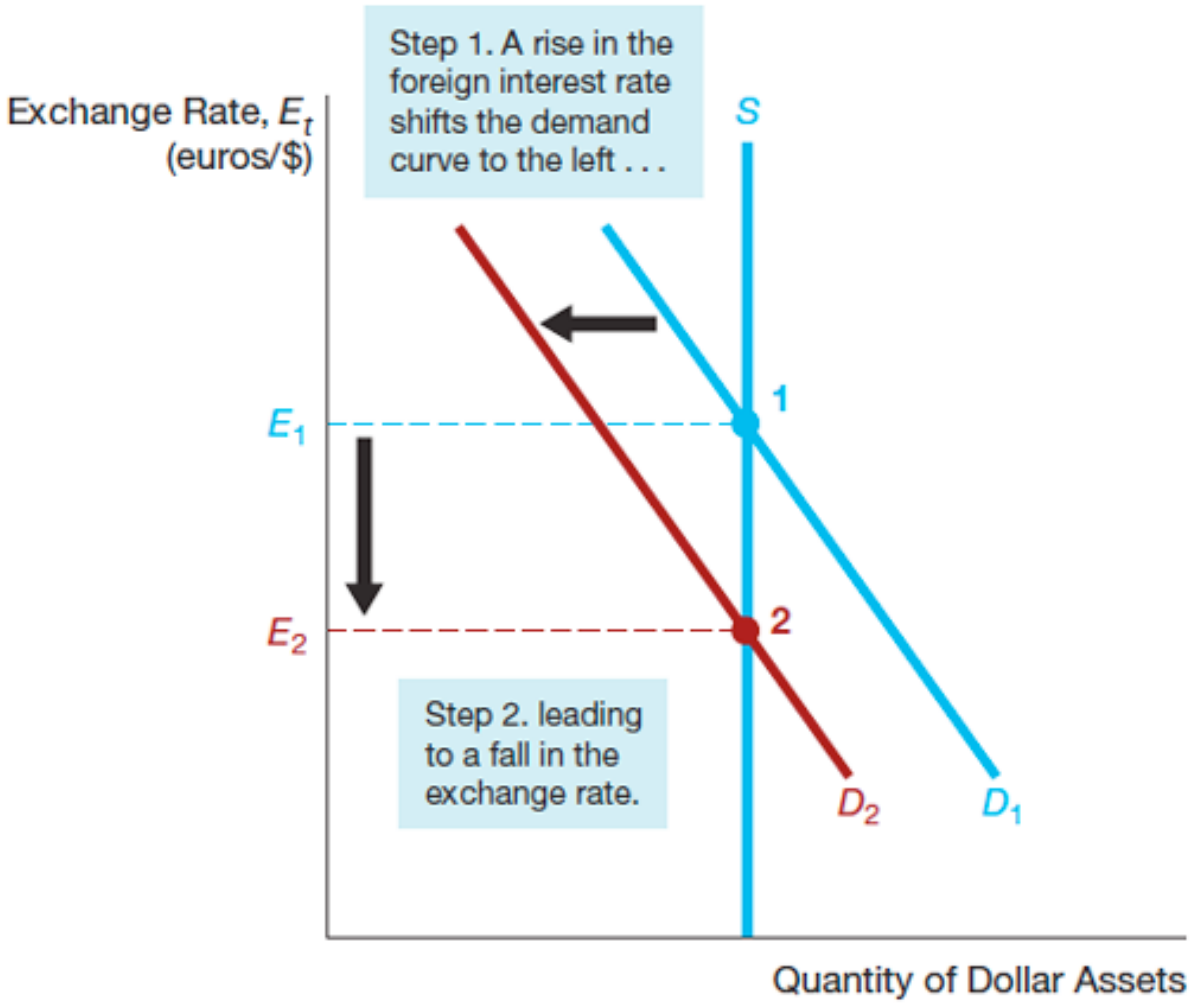

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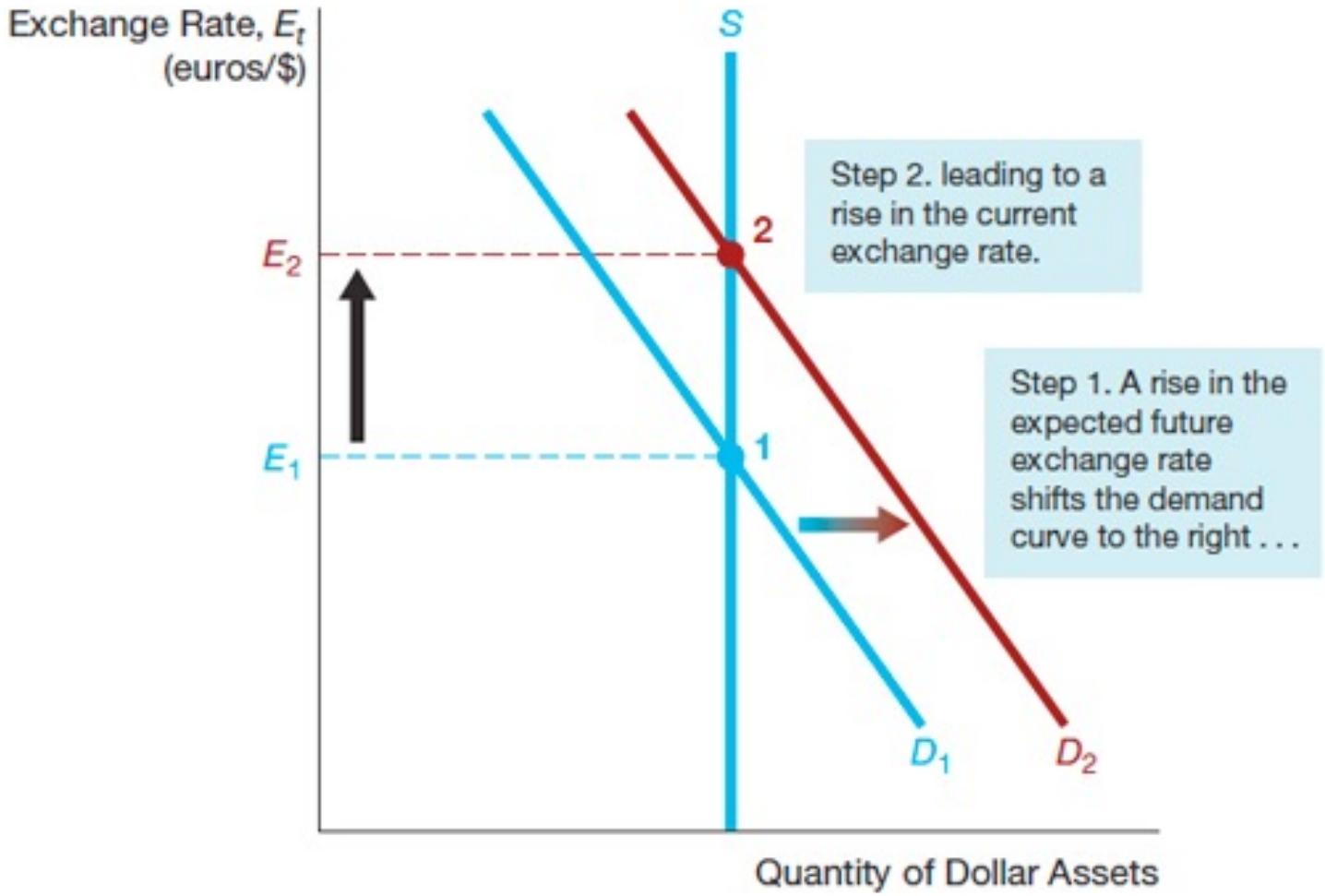
Figure 5 Response to an Increase in the Expected Future Exchange Rate, E^e_{t+1}



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Changes in the Expected Future Exchange Rate

- The following changes, all of which increase the demand for domestic goods relative to foreign goods, will raise the expected future exchange rate
- (1) Expectations of a fall in the American price level relative to the foreign price level;
- (2) Expectations of higher American trade barriers relative to foreign trade barriers;
- (3) Expectations of lower American import demand
- (4) Expectations of higher foreign demand for American exports

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Summary Table 2 Factors That Shift the Demand Curve for Domestic Assets and Affect the Exchange Rate

Factors That Shift the Demand Curve for Domestic Assets and Affect the Exchange Rate

Factor	Change in Factor	Change in Quantity Demanded of Domestic Assets at Each Exchange Rate	Response of Exchange Rate, E_t	
Domestic interest rate, i^D	↑	↑	↑	
Foreign interest rate, i^F	↑	↓	↓	
Expected domestic price level*	↑	↓	↓	
Expected trade barriers*	↑	↑	↑	
Expected import demand	↑	↓	↓	
Expected export demand	↑	↑	↑	
Expected productivity*	↑	↑	↑	

*Relative to other countries.

Note: Only increases (↑) in the factors are shown; the effects of decreases in the variables on the exchange rate are the opposite of those indicated in the "Response" column.

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